

SOLIDWORKS API (NET, VBA, C#) — Reference

(72/337): IDimension

SOLIDWORKS API Documentation (converted from CHM)

IDimension (Type)

IDimension Interface

[See Also](#) [Members](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) : IDimension Interface

Allows you to get and set dimension values and tolerances.

NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

.NET Syntax

Visual Basic (Declaration)

```
Public Interface IDimension
```

Visual Basic (Usage)

```
Dim instance As IDimension
```

C#

```
public interface IDimension
```

C++/CLI

```
public interface class IDimension
```

Visual Basic for Applications (VBA) Syntax

See [Dimension](#).

Example

[Set Dimensions to Mid-tolerance \(VBA\)](#)
[Get and Set Sensor \(C#\)](#)
[Get and Set Sensor \(VB.NET\)](#)
[Get and Set Sensor \(VBA\)](#)

Accessors

[IDisplayDimension2::GetDimension2](#)
[IFeature::Parameter](#) and [IFeature::IParameter](#)
[IModelDoc2::Parameter](#) and [IModelDoc2::IParameter](#)

Access Diagram

[Dimension](#)

See Also

[IDimension Members](#)
[SolidWorks.Interop.sldworks Namespace](#)
[IDisplayDimension Interface](#)
[IDimensionTolerance Interface](#)

IDimension Interface Methods

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : IDimension Interface

For a list of all members of this type, see [IDimension members](#).

Public Methods

Name	Description
 GetArcEndCondition	Gets the end conditions for linear dimensions that end on an arc.
 GetFeatureOwner	Gets the feature for this dimension.
 GetNameForSelection	Gets the name of the selected dimension needed by IModelDocExtension::SelectByID2 .
 GetReferencePointsCount	Gets the number of reference points for this dimension.
 GetSystemChamferValues	Gets the chamfer dimension values in system units.
 GetSystemValue2	Obsolete. See IDimension::GetSystemValue3 and IDimension::IGetSystemValue3 .
 GetSystemValue3	Gets the value of the current dimension in system units in the named configuration.
 GetToleranceFitValues	Obsolete. Superseded by IDimensionTolerance::GetHoleFitValue and IDimensionTolerance::GetShaftFitValue .
 GetToleranceFontInfo	Obsolete. Superseded by IDimensionTolerance::GetFontUseDimension , IDimensionTolerance::GetFontUseScale , IDimensionTolerance::GetFontScale , and IDimensionTolerance::GetFontHeight .
 GetToleranceType	Obsolete. Superseded by IDimensionTolerance::Type .
 GetToleranceValues	Obsolete. Superseded by IDimensionTolerance::GetMaxValue and IDimensionTolerance::GetMinValue .
 GetType	Gets the type of dimension.
 GetUserValueIn	Gets the value of this dimension in the units of the specified document.
 GetValue2	Obsolete. Superseded by IDimension::GetValue3 and IDimension::IGetValue3 .
 GetValue3	Gets the values of the dimensions in the specified configurations.
 IGetReferencePoints	Gets the reference points for this dimension.
 IGetSystemValue3	Gets the value of the current dimension in system units in the named configuration.
 IGetToleranceFontInfo	Obsolete. Superseded by IDimensionTolerance::GetFontUseDimension , IDimensionTolerance::GetFontUseScale , IDimensionTolerance::GetFontScale , and IDimensionTolerance::GetFontHeight .
 IGetToleranceValues	Obsolete. Superseded by IDimensionTolerance::GetMaxValue and IDimensionTolerance::GetMinValue .
 IGetUserValueIn	Obsolete. Superseded by IDimension::IGetUserValueIn2 .
 IGetUserValueIn2	Gets the value of this dimension in the units of the specified document.
 IGetValue3	Gets the values of the dimensions in the specified configurations.
 IsAppliedToAllConfigurations	Gets whether a dimension is currently applied to all configurations of the model or to just the current configuration.
 IsDesignTableDimension	Gets whether this dimension is driven by a design table.
 ISetReferencePoints	Sets the reference points for this dimension.
 ISetSystemValue3	Sets the value of this dimension in system units (meters) in the specified configuration.
 ISetUserValueIn	Obsolete. Superseded by IDimension::ISetUserValueIn3 .
 ISetUserValueIn2	Obsolete. Superseded by IDimension::ISetUserValueIn3 .
 ISetUserValueIn3	Sets the value of this dimension in the units of the specified document.
 ISetValue3	Sets the values of the dimensions in the specified configurations.
 IsReference	Gets whether the dimension is a reference dimension.
 SetArcEndCondition	Sets the end conditions for linear dimensions that end on an arc.
 SetSystemValue2	Obsolete. Superseded by IDimension::SetSystemValue3 .
 SetSystemValue3	Sets the value of this dimension in system units (meters) in the specified configuration.
 SetToleranceFitValues	Obsolete. Superseded by IDimensionTolerance::SetFitValues .
 SetToleranceFontInfo	Obsolete. Superseded by IDimensionTolerance::SetFont .
 SetToleranceType	Obsolete. Superseded by IDimensionTolerance::Type .
 SetToleranceValues	Obsolete. Superseded by IDimensionTolerance::SetValues .
 SetUserValueIn	Obsolete. Superseded by IDimension::SetUserValueIn2 .
 SetUserValueIn2	Sets the value of this dimension in the units of the specified document.
 SetValue2	Obsolete. Superseded by IDimension::SetValue3 .
 SetValue3	Sets the values of the dimensions in the specified configurations.

[Top](#)

See Also

[IDimension Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[IDisplayDimension Interface](#)
[IDimensionTolerance Interface](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IDimension Interface Members

[See Also](#) [Properties](#) [Methods](#)

[SolidWorks.Interop.sldworks Namespace](#) : IDimension Interface

The following tables list the members exposed by [IDimension](#).

Public Properties

Name	Description
 DimensionLineDirection	Gets or sets the direction of this dimension line.
 DrivenState	Gets or sets the driven state of a dimension.
 ExtensionLineDirection	Gets or sets the direction of the extension line.
 FullName	Gets the full name of a dimension including the feature and the model.
 Name	Gets or sets the name of a dimension.
 ReadOnly	Gets or sets the read-only state of a dimension.
 ReferencePoints	Gets or sets the reference points for this dimension.
 SystemValue	Obsolete. Superseded by IDimension::GetSystemValue3 , IDimension::IGetSystemValue3 , IDimension::SetSystemValue3 , and IDimension::ISetSystemValue3 .
 Tolerance	Gets the IDimensionTolerance object.
 Value	Obsolete. Superseded by IDimension::GetValue3 , IDimension::IGetValue3 , IDimension::ISetValue3 , and IDimension::SetValue3 .

[Top](#)

Public Methods

Name	Description
 GetArcEndCondition	Gets the end conditions for linear dimensions that end on an arc.
 GetFeatureOwner	Gets the feature for this dimension.
 GetNameForSelection	Gets the name of the selected dimension needed by IModelDocExtension::SelectByID2 .
 GetReferencePointsCount	Gets the number of reference points for this dimension.
 GetSystemChamferValues	Gets the chamfer dimension values in system units.
 GetSystemValue2	Obsolete. See IDimension::GetSystemValue3 and IDimension::IGetSystemValue3 .
 GetSystemValue3	Gets the value of the current dimension in system units in the named configuration.
 GetToleranceFitValues	Obsolete. Superseded by IDimensionTolerance::GetHoleFitValue and IDimensionTolerance::GetShaftFitValue .
 GetToleranceFontInfo	Obsolete. Superseded by IDimensionTolerance::GetFontUseDimension , IDimensionTolerance::GetFontUseScale , IDimensionTolerance::GetFontScale , and IDimensionTolerance::GetFontHeight .
 GetToleranceType	Obsolete. Superseded by IDimensionTolerance::Type .
 GetToleranceValues	Obsolete. Superseded by IDimensionTolerance::GetMaxValue and IDimensionTolerance::GetMinValue .
 GetType	Gets the type of dimension.
 GetUserValueIn	Gets the value of this dimension in the units of the specified document.
 GetValue2	Obsolete. Superseded by IDimension::GetValue3 and IDimension::IGetValue3 .
 GetValue3	Gets the values of the dimensions in the specified configurations.
 IGetReferencePoints	Gets the reference points for this dimension.
 IGetSystemValue3	Gets the value of the current dimension in system units in the named configuration.
IGetToleranceFontInfo	Obsolete. Superseded by IDimensionTolerance::GetFontUseDimension , IDimensionTolerance::GetFontUseScale , IDimensionTolerance::GetFontScale , and IDimensionTolerance::GetFontHeight .

		
	IGetToleranceValues	Obsolete. Superseded by IDimensionTolerance::GetMaxValue and IDimensionTolerance::GetMinValue
	IGetUserValueIn	Obsolete. Superseded by IDimension::IGetUserValueIn2
	IGetUserValueIn2	Gets the value of this dimension in the units of the specified document.
	IGetValue3	Gets the values of the dimensions in the specified configurations.
	IsAppliedToAllConfigurations	Gets whether a dimension is currently applied to all configurations of the model or to just the current configuration.
	IsDesignTableDimension	Gets whether this dimension is driven by a design table.
	ISetReferencePoints	Sets the reference points for this dimension.
	ISetSystemValue3	Sets the value of this dimension in system units (meters) in the specified configuration.
	ISetUserValueIn	Obsolete. Superseded by IDimension::ISetUserValueIn3
	ISetUserValueIn2	Obsolete. Superseded by IDimension::ISetUserValueIn3
	ISetUserValueIn3	Sets the value of this dimension in the units of the specified document.
	ISetValue3	Sets the values of the dimensions in the specified configurations.
	IsReference	Gets whether the dimension is a reference dimension.
	SetArcEndCondition	Sets the end conditions for linear dimensions that end on an arc.
	SetSystemValue2	Obsolete. Superseded by IDimension::SetSystemValue3
	SetSystemValue3	Sets the value of this dimension in system units (meters) in the specified configuration.
	SetToleranceFitValues	Obsolete. Superseded by IDimensionTolerance::SetFitValues
	SetToleranceFontInfo	Obsolete. Superseded by IDimensionTolerance::SetFont
	SetToleranceType	Obsolete. Superseded by IDimensionTolerance::Type
	SetToleranceValues	Obsolete. Superseded by IDimensionTolerance::SetValues
	SetUserValueIn	Obsolete. Superseded by IDimension::SetUserValueIn2
	SetUserValueIn2	Sets the value of this dimension in the units of the specified document.
	SetValue2	Obsolete. Superseded by IDimension::SetValue3
	SetValue3	Sets the values of the dimensions in the specified configurations.

[Top](#)

See Also

[IDimension Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[IDisplayDimension Interface](#)
[IDimensionTolerance Interface](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IDimension Interface Properties

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : IDimension Interface

For a list of all members of this type, see [IDimension members](#).

Public Properties

Name	Description
 DimensionLineDirection	Gets or sets the direction of this dimension line.
 DrivenState	Gets or sets the driven state of a dimension.
 ExtensionLineDirection	Gets or sets the direction of the extension line.
 FullName	Gets the full name of a dimension including the feature and the model.
 Name	Gets or sets the name of a dimension.
 ReadOnly	Gets or sets the read-only state of a dimension.

	ReferencePoints	Gets or sets the reference points for this dimension.
	SystemValue	Obsolete. Superseded by IDimension::GetSystemValue3 , IDimension::IGetSystemValue3 , IDimension::SetSystemValue3 , and IDimension::ISetSystemValue3 .
	Tolerance	Gets the IDimensionTolerance object.
	Value	Obsolete. Superseded by IDimension::GetValue3 , IDimension::IGetValue3 , IDimension::SetValue3 , and IDimension::ISetValue3 .

[Top](#)

See Also

[IDimension Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[IDisplayDimension Interface](#)
[IDimensionTolerance Interface](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

DimensionLineDirection Property (IDimension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDimension Interface](#) : DimensionLineDirection Property (IDimension)

Gets or sets the direction of this dimension line.

Visual Basic Syntax

Visual Basic (Declaration)

```
Property DimensionLineDirection As MathVector
```

Visual Basic (Usage)



```
Dim instance As IDimension  
Dim value As MathVector
```

```
instance.DimensionLineDirection = value
```

```
value = instance.DimensionLineDirection
```

C#

```
MathVector DimensionLineDirection {get; set;}
```

C++/CLI

```
property MathVector^ DimensionLineDirection {  
    MathVector^ get();  
    void set ( & MathVector^ value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Pointer to the [IMathVector](#) object

Visual Basic for Applications (VBA) Syntax

See [Dimension::DimensionLineDirection](#).

Remarks

This method only supports feature dimensions.

See Also

[IDimension Interface](#)
[IDimension Members](#)
[IDimension::ExtensionLineDirection Property](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

DrivenState Property (IDimension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDimension Interface](#) : DrivenState Property (IDimension)

Gets or sets the driven state of a dimension.

.NET Syntax

Visual Basic (Declaration)

Property DrivenState As System.Integer

Visual Basic (Usage)



```
Dim instance As IDimension
Dim value As System.Integer

instance.DrivenState = value
value = instance.DrivenState
```

C#

```
System.int DrivenState {get; set;}
```

C++/CLI

```
property System.int DrivenState {
    System.int get();
    void set ( & System.int value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Driven state of the dimension as defined in [swDimensionDrivenState_e](#)

Visual Basic for Applications (VBA) Syntax

See [Dimension.DrivenState](#).

Example

[Get Block Information \(VBA\)](#)
[Get Dimension Values in All Configurations \(VBA\)](#)
[Get Dimension Values in Drawing \(VBA\)](#)
[Set Dimensions to Mid-tolerance \(VBA\)](#)

See Also

[IDimension Interface](#)
[IDimension Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

ExtensionLineDirection Property (IDimension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDimension Interface](#) : ExtensionLineDirection Property (IDimension)

Gets or sets the direction of the extension line.

.NET Syntax

Visual Basic (Declaration)

Property ExtensionLineDirection As MathVector

Visual Basic (Usage)



```
Dim instance As IDimension  
Dim value As MathVector  
  
instance.ExtensionLineDirection = value  
  
value = instance.ExtensionLineDirection
```

C#

```
MathVector ExtensionLineDirection {get; set;}
```

C++/CLI

```
property MathVector^ ExtensionLineDirection {  
    MathVector^ get();  
    void set ( & MathVector^ value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Pointer to the [IMathVector](#) object

Visual Basic for Applications (VBA) Syntax

See [Dimension::ExtensionLineDirection](#).

Example

[Get Extension Line Direction \(VBA\)](#)

Remarks

This method only supports feature dimensions. Additionally, it returns non-0 vector for feature dimensions ([IDimension::GetFeatureOwner](#)) and 0 vector for radial and chamfer dimensions.

See Also

[IDimension Interface](#)
[IDimension Members](#)
[IDimension::DimensionLineDirection Property](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

FullName Property (IDimension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDimension Interface](#) : FullName Property (IDimension)

Gets the full name of a dimension including the feature and the model.

.NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property FullName As System.String
```

Visual Basic (Usage)



```
Dim instance As IDimension  
Dim value As System.String
```

```
value = instance.FullName
```

C#

```
System.string FullName {get;}
```

C++/CLI

```
property System.String^ FullName {  
    System.String^ get();  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Name of the dimension including the feature and model

Visual Basic for Applications (VBA) Syntax

See [Dimension::FullName](#).

Example

[Determine if Display Dimension Marked for Drawing \(VBA\)](#)
[Get Block Information \(VBA\)](#)
[Get Dimension Values in All Configurations \(VBA\)](#)
[Get Mate Definition \(VBA\)](#)
[Iterate Through Dimensions in Model \(VBA\)](#)
[Set Dimensions to Mid-tolerance \(VBA\)](#)

Remarks

The syntax of the name returned is:

```
<Dimension Name>@<Feature Name>@<Model>
```

where:

- <Dimension Name> is the name of the dimension as returned from [IDimension::Name](#)
- <Feature Name> is the name of the feature that the dimension is on
- <Model> is the name of the model containing the feature

For example:

```
"D1@Sketch1@Part4.Part"
```

See Also

[IDimension Interface](#)
[IDimension Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetArcEndCondition Method (IDimension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDimension Interface](#) : GetArcEndCondition Method (IDimension)

Gets the end conditions for linear dimensions that end on an arc.

.NET Syntax

Visual Basic (Declaration)

```
Function GetArcEndCondition( _  
    ByVal Index As System.Integer _  
) As System.Integer
```

Visual Basic (Usage)

 Copy Code

```
Dim instance As IDimension
Dim Index As System.Integer
Dim value As System.Integer

value = instance.GetArcEndCondition(Index)
```

C#

```
System.int GetArcEndCondition(
    System.int Index
)
```

C++/CLI

```
System.int GetArcEndCondition(
    & System.int Index
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index

Index of the endpoint to get the end condition from; 1 is the first endpoint, 2 is the second endpoint

Return Value

End condition as defined in [swArcEndCondition_e](#)

Visual Basic for Applications (VBA) Syntax

See [Dimension::GetArcEndCondition](#).

Remarks

In a linear dimension, the distance measured is from one point to another. If one or both of those points are on an arc, you can change the point to be the center of the arc, the nearest point on the arc, or the furthest point on the arc. The arc end condition describes which point is used.

Use [IDimension::SetArcEndCondition](#) to set the arc end conditions.

See Also

[IDimension Interface](#)
[IDimension Members](#)
[IDimension::SetArcEndCondition Method](#)

Availability

SOLIDWORKS 99, datecode 1999207

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetFeatureOwner Method (IDimension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDimension Interface](#) : GetFeatureOwner Method (IDimension)

Gets the feature for this dimension.

.NET Syntax

Visual Basic (Declaration)

```
Function GetFeatureOwner() As Feature
```

Visual Basic (Usage)

```
Dim instance As IDimension
Dim value As Feature
```



```
value = instance.GetFeatureOwner()
```

C#

```
Feature GetFeatureOwner()
```

C++/CLI

```
Feature^ GetFeatureOwner();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Pointer to the [IFeature](#) object for this dimension

Visual Basic for Applications (VBA) Syntax

See [Dimension::GetFeatureOwner](#).

Example

[Get Component Via Display Dimension \(VBA\)](#)

See Also

[IDimension Interface](#)
[IDimension Members](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetNameForSelection Method (IDimension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDimension Interface](#) : GetNameForSelection Method (IDimension)

Gets the name of the selected dimension needed by [IModelDocExtension::SelectByID2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetNameForSelection() As System.String
```

Visual Basic (Usage)



```
Dim instance As IDimension  
Dim value As System.String  
  
value = instance.GetNameForSelection()
```

C#

```
System.string GetNameForSelection()
```

C++/CLI

```
System.String^ GetNameForSelection();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Name of selected dimension

Visual Basic for Applications (VBA) Syntax

See [Dimension::GetNameForSelection](#).

See Also

[IDimension Interface](#)
[IDimension Members](#)
[IDisplayDimension::GetNameForSelection Method](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetReferencePointsCount Method (IDimension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDimension Interface](#) : GetReferencePointsCount Method (IDimension)

Gets the number of reference points for this dimension.

.NET Syntax

Visual Basic (Declaration)

```
Function GetReferencePointsCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IDimension  
Dim value As System.Integer  
  
value = instance.GetReferencePointsCount()
```

C#

```
System.int GetReferencePointsCount()
```

C++/CLI

```
System.int GetReferencePointsCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of dimension reference points

Visual Basic for Applications (VBA) Syntax

See [Dimension::GetReferencePointsCount](#).

Remarks

Call this method before calling [IDimension::IGetReferencePoints](#).

See Also

[IDimension Interface](#)
[IDimension Members](#)
[IDimension::ISetReferencePoints Method](#)
[IDimension::ReferencePoints Property](#)

Availability

SOLIDWORKS 2004 Revision Number 12.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetSystemChamferValues Method (IDimension)

[See Also](#) [Example](#)



Gets the chamfer dimension values in system units.

Visual Basic Syntax

Visual Basic (Declaration)

```
Function GetSystemChamferValues( _  
    ByRef Length As System.Double, _  
    ByRef Angle As System.Double _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IDimension  
Dim Length As System.Double  
Dim Angle As System.Double  
Dim value As System.Boolean
```

```
value = instance.GetSystemChamferValues(Length, Angle)
```

C#

```
System.bool GetSystemChamferValues(  
    out System.double Length,  
    out System.double Angle  
)
```

C++/CLI

```
System.bool GetSystemChamferValues(  
& [Out] System.double Length,  
& [Out] System.double Angle  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Length
Length of chamfer
Angle
Angle of chamfer

Return Value

True if the dimension is a chamfer dimension, false if not

Visual Basic for Applications (VBA) Syntax

See [Dimension::GetSystemChamferValues](#).

Example

[Get Chamfer Dimension \(C#\)](#)
[Get Chamfer Dimension \(VB.NET\)](#)
[Get Chamfer Dimension \(VBA\)](#)

Remarks

Unlike most other types of dimensions, the values returned for a chamfer dimension are not necessarily the values seen by the user in the displayed dimension text. The display dimension interprets these values and considers the type of chamfer display requested by the user and then creates an appropriate output string.

See Also

[IDimension Interface](#)
[IDimension Members](#)
[IDrawingDoc::AddChamferDim Method](#)
[IDrawingDoc::IAddChamferDim Method](#)

Availability

SOLIDWORKS 2006 SP1, Revision Number 14.1

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetSystemValue2 Method (IDimension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDimension Interface](#) : GetSystemValue2 Method (IDimension)

Obsolete. See [IDimension::GetSystemValue3](#) and [IDimension::IGetSystemValue3](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetSystemValue2( _  
    ByVal ConfigName As System.String _  
) As System.Double
```

Visual Basic (Usage)



```
Dim instance As IDimension  
Dim ConfigName As System.String  
Dim value As System.Double  
  
value = instance.GetSystemValue2(ConfigName)
```

C#

```
System.double GetSystemValue2(  
    System.string ConfigName  
)
```

C++/CLI

```
System.double GetSystemValue2(  
& System.String^ ConfigName  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ConfigName
Name of configuration

Return Value

Value in system units

Visual Basic for Applications (VBA) Syntax

See [Dimension::GetSystemValue2](#).

See Also

[IDimension Interface](#)
[IDimension Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetSystemValue3 Method (IDimension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDimension Interface](#) : GetSystemValue3 Method (IDimension)

Gets the value of the current dimension in system units in the named configuration.

.NET Syntax

Visual Basic (Declaration)

```
Function GetSystemValue3( _
    ByVal WhichConfigurations As System.Integer, _
    ByVal Config\_names As System.Object _
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IDimension
Dim WhichConfigurations As System.Integer
Dim Config_names As System.Object
Dim value As System.Object

value = instance.GetSystemValue3(WhichConfigurations, Config_names)
```

C#

```
System.object GetSystemValue3(
    System.int WhichConfigurations,
    System.object Config\_names
)
```

C++/CLI

```
System.Object^ GetSystemValue3(
    & System.int WhichConfigurations,
    & System.Object^ Config\_names
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

WhichConfigurations
Configurations to get this value from as defined by [swInConfigurationOpts_e](#) (see **Remarks**)

Config_names
Names of the configurations (see **Remarks**)

Return Value

Array of doubles of the dimension values in system units

Visual Basic for Applications (VBA) Syntax

See [Dimension::GetSystemValue3](#).

Example

[Change Dimensions of Gear Mate \(VBA\)](#)
[Edit Mate \(VBA\)](#)
[Get Dimension of Distance Mate \(VBA\)](#)
[Get Dimension Values in All Configurations \(VBA\)](#)
[Get Mate Definition \(VBA\)](#)
[Modify Plane By Changing System Value \(VBA\)](#)
[Get Depth of Extrusion \(VBA\)](#)
[Get Depth of Extrusion \(VB.NET\)](#)
[Get Depth of Extrusion \(C#\)](#)

Remarks

The Config_names argument is only used if WhichConfigurations is set to swSpecifyConfiguration.

In OLE, Config_names can be either a BSTR array or a single BSTR.

See Also

[IDimension Interface](#)
[IDimension Members](#)
[IDimension::IGetSystemValue3 Method](#)
[IDimension::ISetSystemValue3 Method](#)
[IDimension::SetSystemValue3 Method](#)
[IDimension::SetUserValueIn2 Method](#)
[IDimension::SetUserValueIn3 Method](#)
[IDimension::GetUserValueIn Method](#)
[IDimension::IGetUserValueIn2 Method](#)
[IDimension::GetValue3 Method](#)

[IDimension::GetValue3 Method](#)
[IDimension::SetValue3 Method](#)
[IDimension::SetValue3 Method](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetToleranceFitValues Method (IDimension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDimension Interface](#) : GetToleranceFitValues Method (IDimension)

Obsolete. Superseded by [IDimensionTolerance::GetHoleFitValue](#) and [IDimensionTolerance::GetShaftFitValue](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetToleranceFitValues() As System.String
```

Visual Basic (Usage)



```
Dim instance As IDimension  
Dim value As System.String  
  
value = instance.GetToleranceFitValues()
```

C#

```
System.string GetToleranceFitValues()
```

C++/CLI

```
System.String^ GetToleranceFitValues();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [Dimension::GetToleranceFitValues](#).

See Also

[IDimension Interface](#)
[IDimension Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetToleranceFontInfo Method (IDimension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDimension Interface](#) : GetToleranceFontInfo Method (IDimension)

Obsolete. Superseded by [IDimensionTolerance::GetFontUseDimension](#), [IDimensionTolerance::GetFontUseScale](#), [IDimensionTolerance::GetFontScale](#), and [IDimensionTolerance::GetFontHeight](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetToleranceFontInfo() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IDimension  
Dim value As System.Object  
  
value = instance.GetToleranceFontInfo()
```

C#

```
System.Object GetToleranceFontInfo()
```

C++/CLI

```
System.Object^ GetToleranceFontInfo();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [Dimension::GetToleranceFontInfo](#).

See Also

[IDimension Interface](#)
[IDimension Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetToleranceType Method (IDimension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDimension Interface](#) : GetToleranceType Method (IDimension)

Obsolete. Superseded by [IDimensionTolerance::Type](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetToleranceType() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IDimension  
Dim value As System.Integer  
  
value = instance.GetToleranceType()
```

C#

```
System.int GetToleranceType()
```

C++/CLI

```
System.int GetToleranceType();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [Dimension::GetToleranceType](#).

See Also

[IDimension Interface](#)
[IDimension Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetToleranceValues Method (IDimension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDimension Interface](#) : GetToleranceValues Method (IDimension)

Obsolete. Superseded by [IDimensionTolerance::GetMaxValue](#) and [IDimensionTolerance::GetMinValue](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetToleranceValues() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IDimension  
Dim value As System.Object  
  
value = instance.GetToleranceValues()
```

C#

```
System.object GetToleranceValues()
```

C++/CLI

```
System.Object^ GetToleranceValues();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [Dimension::GetToleranceValues](#).

See Also

[IDimension Interface](#)
[IDimension Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetType Method (IDimension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDimension Interface](#) : GetType Method (IDimension)

Gets the type of dimension.

.NET Syntax

Visual Basic (Declaration)

```
Function GetType() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IDimension  
Dim value As System.Integer  
  
value = instance.GetType()
```

C#

```
System.int GetType()
```

C++/CLI

```
System.int GetType();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Type of dimension as defined by [swDimensionParamType_e](#)

Visual Basic for Applications (VBA) Syntax

See [Dimension::GetType](#).

Example

[Get Block Information \(VBA\)](#)
[Set Dimensions to Mid-tolerance \(VBA\)](#)
[Get Dimension Values in Drawing \(VBA\)](#)
[Get Dimension Values in Drawing \(VB.NET\)](#)
[Get Dimension Values in Drawing \(C#\)](#)

See Also

[IDimension Interface](#)
[IDimension Members](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetUserValueIn Method (IDimension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDimension Interface](#) : GetUserValueIn Method (IDimension)

Gets the value of this dimension in the units of the specified document.

.NET Syntax

Visual Basic (Declaration)

```
Function GetUserValueIn(  
    ByVal Doc As System.Object _  
) As System.Double
```

Visual Basic (Usage)



```
Dim instance As IDimension  
Dim Doc As System.Object  
Dim value As System.Double  
  
value = instance.GetUserValueIn(Doc)
```

C#

```
System.double GetUserValueIn(  
    System.object Doc  
)
```

C++/CLI

```
System.double GetUserValueIn(  
& System.Object^ Doc  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Doc
Document whose units you want to use

Return Value

Value of this dimension in the units of the specified document, Doc

Visual Basic for Applications (VBA) Syntax

See [Dimension::GetUserValueIn](#).

See Also

[IDimension Interface](#)
[IDimension Members](#)
[IDimension::!GetUserValueIn2 Method](#)
[IDimension::GetSystemValue3 Method](#)
[IDimension::!GetSystemValue3 Method](#)
[IDimension::!SetSystemValue3 Method](#)
[IDimension::SetSystemValue3 Method](#)
[IDimension::SetUserValueIn2 Method](#)
[IDimension::GetValue3 Method](#)
[IDimension::!GetValue3 Method](#)
[IDimension::!SetUserValueIn3 Method](#)
[IDimension::!SetValue3 Method](#)
[IDimension::SetValue3 Method](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetValue2 Method (IDimension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDimension Interface](#) : GetValue2 Method (IDimension)

Obsolete. Superseded by [IDimension::GetValue3](#) and [IDimension::!GetValue3](#).

.NET Syntax

Visual Basic (Declaration)

```
Function GetValue2( _  
    ByVal ConfigName As System.String _  
) As System.Double
```

Visual Basic (Usage)



```
Dim instance As IDimension  
Dim ConfigName As System.String  
Dim value As System.Double  
  
value = instance.GetValue2(ConfigName)
```

C#

```
System.double GetValue2(  
    System.string ConfigName  
)
```

C++/CLI

```
System.double GetValue2(  
& System.String^ ConfigName  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ConfigName

Visual Basic for Applications (VBA) Syntax

See [Dimension::GetValue2](#).

See Also

[IDimension Interface](#)
[IDimension Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetValue3 Method (IDimension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDimension Interface](#) : GetValue3 Method (IDimension)

Gets the values of the dimensions in the specified configurations.

.NET Syntax

```
Function GetValue3( _
    ByVal WhichConfigurations As System.Integer, _
    ByVal Config\_names As System.Object _
) As System.Object
```

 Copy Code

```
Dim instance As IDimension
Dim WhichConfigurations As System.Integer
Dim Config_names As System.Object
Dim value As System.Object

value = instance.GetValue3(WhichConfigurations, Config_names)
```

```
System.object GetValue3(
    System.int WhichConfigurations,
    System.object Config_names
)
```

```
System.Object^ GetValue3(
&    System.int WhichConfigurations,
&    System.Object^ Config_names
)
```

Parameters

Names of the configurations (see **Remarks**)

Array of doubles of the dimensions for the specified configurations

See [Dimension::GetValue3](#).

[Get Dimension Values in All Configurations \(VBA\).](#)

The `Config_names` argument is only used if `WhichConfigurations` is set to `swSpecifyConfiguration`. `Config_names` can contain either a BSTR array or a single BSTR.

- [IDimension Interface](#)
- [IDimension Members](#)
- [IDimension::IGetValue3 Method](#)
- [IDimension::SetValue3 Method](#)
- [IDimension::GetUserValueIn Method](#)
- [IDimension::GetSystemValue3 Method](#)
- [IDimension::IGetSystemValue3 Method](#)
- [IDimension::IGetUserValueIn2 Method](#)
- [IDimension::ISetUserValueIn3 Method](#)
- [IDimension::ISetValue3 Method](#)
- [IDimension::SetSystemValue3 Method](#)
- [IDimension::SetValue3 Method](#)
- [IDimension::SetUserValueIn2 Method](#)
- [IDimension::ISetSystemValue3 Method](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetReferencePoints Method (IDimension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDimension Interface](#) : IGetReferencePoints Method (IDimension)

Gets the reference points for this dimension.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetReferencePoints( _  
    ByVal PointsCount As System.Integer _  
) As MathPoint
```

Visual Basic (Usage)



```
Dim instance As IDimension  
Dim PointsCount As System.Integer  
Dim value As MathPoint  
  
value = instance.IGetReferencePoints(PointsCount)
```

C#

```
MathPoint IGetReferencePoints(  
    System.int PointsCount  
)
```

C++/CLI

```
MathPoint^ IGetReferencePoints(  
&    System.int PointsCount  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

PointsCount

Number of reference points for this dimension

Return Value

- in-process, unmanaged C++: Pointer to the [IMathPoint](#) object
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

Call [IDimension::IGetReferencePointsCount](#) before calling this method to determine the size of the array.

See Also

[IDimension Interface](#)
[IDimension Members](#)
[IDimension::ISetReferencePoints Method](#)
[IDimension::ReferencePoints Property](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetSystemValue3 Method (IDimension)

[See Also](#)



Gets the value of the current dimension in system units in the named configuration.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetSystemValue3( _  
    ByVal WhichConfigurations As System.Integer, _  
    ByVal Config\_count As System.Integer, _  
    ByRef Config\_names As System.String _  
) As System.Double
```

Visual Basic (Usage)



```
Dim instance As IDimension  
Dim WhichConfigurations As System.Integer  
Dim Config_count As System.Integer  
Dim Config_names As System.String  
Dim value As System.Double  
  
value = instance.IGetSystemValue3(WhichConfigurations, Config_count, Config_names)
```

C#

```
System.double IGetSystemValue3(  
    System.int WhichConfigurations,  
    System.int Config\_count,  
    ref System.string Config\_names  
)
```

C++/CLI

```
System.double IGetSystemValue3(  
& System.int WhichConfigurations,  
& System.int Config\_count,  
& System.String^% Config\_names  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

WhichConfigurations

Configurations to get this value from as defined by [swInConfigurationOpts_e](#) (see **Remarks**)

Config_count

Number of configurations (see **Remarks**)

Config_names

Names of the configuration (see **Remarks**)

Return Value

Value in system units

Visual Basic for Applications (VBA) Syntax

See [Dimension::IGetSystemValue3](#).

Remarks

The Config_count and Config_names arguments are only used if WhichConfigurations is set to swSpecifyConfiguration.

See Also

[IDimension Interface](#)
[IDimension Members](#)
[IDimension::GetSystemValue3 Method](#)
[IDimension::IGetUserValueIn2 Method](#)
[IDimension::SetSystemValue3 Method](#)
[IDimension::SetUserValueIn2 Method](#)

[IDimension::!SetSystemValue3 Method](#)
[IDimension::GetUserValueIn Method](#)
[IDimension::GetValue3 Method](#)
[IDimension::!GetValue3 Method](#)
[IDimension::!SetValue3 Method](#)
[IDimension::SetValue3 Method](#)
[IDimension::!SetUserValueIn3 Method](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetToleranceFontInfo Method (IDimension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDimension Interface](#) : IGetToleranceFontInfo Method (IDimension)

Obsolete. Superseded by [IDimensionTolerance::GetFontUseDimension](#), [IDimensionTolerance::GetFontUseScale](#), [IDimensionTolerance::GetFontScale](#), and [IDimensionTolerance::GetFontHeight](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetToleranceFontInfo() As System.Double
```

Visual Basic (Usage)



```
Dim instance As IDimension  
Dim value As System.Double  
  
value = instance.IGetToleranceFontInfo()
```

C#

```
System.double IGetToleranceFontInfo()
```

C++/CLI

```
System.double IGetToleranceFontInfo();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [Dimension::IGetToleranceFontInfo](#).

See Also

[IDimension Interface](#)
[IDimension Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IGetToleranceValues Method (IDimension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDimension Interface](#) : IGetToleranceValues Method (IDimension)

Obsolete. Superseded by [IDimensionTolerance::GetMaxValue](#) and [IDimensionTolerance::GetMinValue](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetToleranceValues() As System.Double
```

Visual Basic (Usage)



```
Dim instance As IDimension  
Dim value As System.Double
```

```
value = instance.IGetToleranceValues()
```

C#

```
System.double IGetToleranceValues()
```

C++/CLI

```
System.double IGetToleranceValues();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [Dimension::IGetToleranceValues](#).

See Also

[IDimension Interface](#)
[IDimension Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IGetUserValueIn Method (IDimension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDimension Interface](#) : IGetUserValueIn Method (IDimension)

Obsolete. Superseded by [IDimension::IGetUserValueIn2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function IGetUserValueIn( _  
    ByVal Doc As ModelDoc _  
) As System.Double
```

Visual Basic (Usage)



```
Dim instance As IDimension  
Dim Doc As ModelDoc  
Dim value As System.Double  
  
value = instance.IGetUserValueIn(Doc)
```

C#

```
System.double IGetUserValueIn(  
    ModelDoc Doc  
)
```

C++/CLI

```
System.double IGetUserValueIn(  
& ModelDoc^ Doc  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Doc

Visual Basic for Applications (VBA) Syntax

See [Dimension::IGetUserValueIn](#).

See Also

IDimension Interface

IDimension Members

SOLIDWORKS API Help

[Send comments](#) on this topic.

IGetValueIn2 Method (IDimension)

See Also



[SolidWorks.Interop.sldworks Namespace](#) > [IDimension Interface](#) : IGetUserValueIn2 Method (IDimension)

Gets the value of this dimension in the units of the specified document.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetUserValueIn2( _  
    ByVal Doc As ModelDoc2 _  
    ) As System.Double
```

Visual Basic (Usage)



```
Dim instance As IDimension  
Dim Doc As ModelDoc2  
Dim value As System.Double
```

```
value = instance.IGetUserValueIn2(Doc)
```

C#

```
System.double IGetUserValueIn2(
    ModelDoc2 Doc
)
```

C++/CLI

```
System.double IGetUserValueIn2(
& ModelDoc2^ Doc
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Doc

Document whose units you want to use

Return Value

Value of this dimension in the units of the document specified by Doc

Visual Basic for Applications (VBA) Syntax

See [Dimension::IGetUserValueIn2](#).

See Also

[IDimension Interface](#)
[IDimension Members](#)
[IDimension::GetUserValueIn Method](#)
[IDimension::GetSystemValue3 Method](#)
[IDimension::IGetSystemValue3 Method](#)
[IDimension::ISetSystemValue3 Method](#)
[IDimension::ISetUserValueIn3 Method](#)
[IDimension::SetSystemValue3 Method](#)
[IDimension::SetUserValueIn2 Method](#)
[IDimension::GetValue3 Method](#)
[IDimension::IGetValue3 Method](#)

[-] Availability

IGetValue3 Method (IDimension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDimension Interface](#) : IGetValue3 Method (IDimension)

Gets the values of the dimensions in the specified configurations.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetValue3( _
    ByVal WhichConfigurations As System.Integer, _
    ByVal Config\_count As System.Integer, _
    ByRef Config\_names As System.String _
) As System.Double
```

Visual Basic (Usage)



```
Dim instance As IDimension
Dim WhichConfigurations As System.Integer
Dim Config_count As System.Integer
Dim Config_names As System.String
Dim value As System.Double

value = instance.IGetValue3(WhichConfigurations, Config_count, Config_names)
```

C#

```
System.double IGetValue3(
    System.int WhichConfigurations,
    System.int Config\_count,
    ref System.string Config\_names
)
```

C++/CLI

```
System.double IGetValue3(
    & System.int WhichConfigurations,
    & System.int Config\_count,
    & System.String^% Config\_names
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

WhichConfigurations

Configurations in which to get this value as defined in [swInConfigurationOpts_e](#) (see **Remarks**)

Config_count

Number of configurations (see **Remarks**)

Config_names

- in-process, unmanaged C++: Pointer to an array of the names of the configurations of size Config_count (see **Remarks**)

- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

Return Value

- in-process, unmanaged C++: Pointer to an array of values of the dimensions for the specified configurations of size Config_count (see **Remarks**)

- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

Remarks

The Config_count and Config_names arguments are only used if WhichConfigurations is set to swSpecifyConfiguration.

See Also

[IDimension Interface](#)

[IDimension Members](#)

[IDimension::SetValue3 Method](#)

[IDimension::GetValue3 Method](#)

[IDimension::GetSystemValue3 Method](#)
[IDimension::GetUserValueIn Method](#)
[IDimension::!GetSystemValue3 Method](#)
[IDimension::!GetUserValueIn2 Method](#)
[IDimension::!GetSystemValue3 Method](#)
[IDimension::!SetSystemValueIn3 Method](#)
[IDimension::!SetUserValueIn3 Method](#)
[IDimension::!SetValue3 Method](#)
[IDimension::SetSystemValue3 Method](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IsAppliedToAllConfigurations Method (IDimension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDimension Interface](#) : IsAppliedToAllConfigurations Method (IDimension)

Gets whether a dimension is currently applied to all configurations of the model or to just the current configuration.

.NET Syntax

Visual Basic (Declaration)

```
Function IsAppliedToAllConfigurations() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IDimension  
Dim value As System.Boolean  
  
value = instance.IsAppliedToAllConfigurations()
```

C#

```
System.bool IsAppliedToAllConfigurations()
```

C++/CLI

```
System.bool IsAppliedToAllConfigurations();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if the dimension is applied to all configurations, false if the dimension is applied only to the current configuration

Visual Basic for Applications (VBA) Syntax

See [Dimension::IsAppliedToAllConfigurations](#).

Example

[Get Dimension Values in All Configurations \(VBA\)](#)

Remarks

If there is only one configuration of a part, this method returns True.

This method applies only to dimensions that are driven by or drive geometry. For example, this method does not apply to reference dimensions (see [IDimension::IsReference](#) and [IDimension::DrivenState](#)). This method returns True for reference dimensions.

See Also

[IDimension Interface](#)
[IDimension Members](#)

Availability

IsDesignTableDimension Method (IDimension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDimension Interface](#) : IsDesignTableDimension Method (IDimension)

Gets whether this dimension is driven by a design table.

.NET Syntax

Visual Basic (Declaration)

```
Function IsDesignTableDimension() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IDimension  
Dim value As System.Boolean  
  
value = instance.IsDesignTableDimension()
```

C#

```
System.bool IsDesignTableDimension()
```

C++/CLI

```
System.bool IsDesignTableDimension();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if the dimension is driven by a design table, false if not

Visual Basic for Applications (VBA) Syntax

See [Dimension::IsDesignTableDimension](#).

See Also

[IDimension Interface](#)
[IDimension Members](#)
[IDimension::DrivenState Property](#)
[IDesignTable Interface](#)

Availability

SOLIDWORKS 2008 SP2, Revision Number 16.2

ISetReferencePoints Method (IDimension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDimension Interface](#) : ISetReferencePoints Method (IDimension)

Sets the reference points for this dimension.

.NET Syntax

Visual Basic (Declaration)

```
Sub ISetReferencePoints( _  
    ByVal PointsCount As System.Integer, _  
    ByRef RefPoints As MathPoint _  
)
```

Visual Basic (Usage)



```
Dim instance As IDimension
Dim PointsCount As System.Integer
Dim RefPoints As MathPoint

instance.ISetReferencePoints(PointsCount, RefPoints)
```

C#

```
void ISetReferencePoints(
    System.int PointsCount,
    ref MathPoint RefPoints
)
```

C++/CLI

```
void ISetReferencePoints(
    & System.int PointsCount,
    & MathPoint^% RefPoints
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

PointsCount
Number of reference points for this dimension
RefPoints
Pointer to the [IMathPoint](#) object

Visual Basic for Applications (VBA) Syntax

See [Dimension::ISetReferencePoints](#).

See Also

[IDimension Interface](#)
[IDimension Members](#)
[IDimension::IGetReferencePoints Method](#)
[IDimension::ReferencePoints Property](#)
[IDimension::GetReferencePointsCount Method](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ISetSystemValue3 Method (IDimension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDimension Interface](#) : ISetSystemValue3 Method (IDimension)

Sets the value of this dimension in system units (meters) in the specified configuration.

.NET Syntax

Visual Basic (Declaration)

```
Function ISetSystemValue3( _
    ByVal NewValue As System.Double, _
    ByVal WhichConfigurations As System.Integer, _
    ByVal Config\_count As System.Integer, _
    ByRef Config\_names As System.String _
) As System.Integer
```

Visual Basic (Usage)

```
Dim instance As IDimension
Dim NewValue As System.Double
Dim WhichConfigurations As System.Integer
Dim Config_count As System.Integer
```



```
Dim Config_names As System.String
Dim value As System.Integer
```

```
value = instance.ISetSystemValue3(NewValue, WhichConfigurations, Config_count, Config_names)
```

C#

```
System.int ISetSystemValue3(
    System.double NewValue,
    System.int WhichConfigurations,
    System.int Config\_count,
    ref System.string Config\_names
)
```

C++/CLI

```
System.int ISetSystemValue3(
    & System.double NewValue,
    & System.int WhichConfigurations,
    & System.int Config\_count,
    & System.String^% Config\_names
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

NewValue

Dimension value in meters

WhichConfigurations

Configuration in which to set this value as defined in [swSetValueInConfiguration_e](#) (see **Remarks**)

Config_count

Names of the configurations (see **Remarks**)

Config_names

Names of the configurations (see **Remarks**)

Return Value

Success indicator value as defined in [swSetValueReturnStatus_e](#)

Visual Basic for Applications (VBA) Syntax

See [Dimension::ISetSystemValue3](#).

Remarks

The WhichConfigurations argument is equivalent to the **Change Parameter** dialog in the SOLIDWORKS Design user interface, which gives the user the option of having the value set in all configurations or the current configuration. If there is one configuration in the part, SOLIDWORKS ignores this argument.

The Config_count and Config_names arguments are only used if WhichConfigurations is set to swSetValue_InSpecificConfigurations.

Config_names can contain either a BSTR array or a single BSTR.

This method allows you to change the value of a read-only dimension. You can use [IDimension::ReadOnly](#) to determine if a dimension is read-only.

See Also

[IDimension Interface](#)
[IDimension Members](#)
[IDimension::SetSystemValue3 Method](#)
[IDimension::GetSystemValue3 Method](#)
[IDimension::GetValue3 Method](#)
[IDimension::ISetValue3 Method](#)
[IDimension::SetValue3 Method](#)
[IDimension::GetUserValueIn Method](#)
[IDimension::ISetUserValueIn2 Method](#)
[IDimension::ISetUserValueIn Method](#)
[IDimension::SetUserValueIn2 Method](#)
[IDimension::GetUserValueIn Method](#)

Obsolete. Superseded by [IDimension::ISetUserValueIn3](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub ISetUserValueIn( _
    ByVal Doc As ModelDoc, _
    ByVal NewValue As System.Double _
)
```

Visual Basic (Usage)



```
Dim instance As IDimension
Dim Doc As ModelDoc
Dim NewValue As System.Double

instance.ISetUserValueIn(Doc, NewValue)
```

C#

```
void ISetUserValueIn(
    ModelDoc Doc,
    System.double NewValue
)
```

C++/CLI

```
void ISetUserValueIn(
    & ModelDoc^ Doc,
    & System.double NewValue
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Doc

NewValue

Visual Basic for Applications (VBA) Syntax

See [Dimension::ISetUserValueIn](#).

See Also

[IDimension Interface](#)
[IDimension Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

ISetUserValueIn2 Method (IDimension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDimension Interface](#) : ISetUserValueIn2 Method (IDimension)

Obsolete. Superseded by [IDimension::ISetUserValueIn3](#).

.NET Syntax

Visual Basic (Declaration)

```
Function ISetUserValueIn2( _
    ByVal Doc As ModelDoc, _
    ByVal NewValue As System.Double, _
    ByVal WhichConfigurations As System.Integer _
) As System.Integer
```

Visual Basic (Usage)

```
Dim instance As IDimension  
Dim Doc As ModelDoc  
Dim NewValue As System.Double  
Dim WhichConfigurations As System.Integer  
Dim value As System.Integer  
  
value = instance.ISetUserValueIn2(Doc, NewValue, WhichConfigurations)
```

C#

```
System.int ISetUserValueIn2(  
    ModelDoc Doc,  
    System.double NewValue,  
    System.int WhichConfigurations  
)
```

C++/CLI

```
System.int ISetUserValueIn2(  
& ModelDoc^ Doc,  
& System.double NewValue,  
& System.int WhichConfigurations  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Doc

NewValue

WhichConfigurations

Visual Basic for Applications (VBA) Syntax

See [Dimension.ISetUserValueIn2](#).

See Also

[IDimension Interface](#)
[IDimension Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

ISetUserValueIn3 Method (IDimension)

[See Also](#)

[SolidWorks.Interop.sldworks Namespace](#) > [IDimension Interface](#) : ISetUserValueIn3 Method (IDimension)

Sets the value of this dimension in the units of the specified document.

.NET Syntax

Visual Basic (Declaration)

```
Function ISetUserValueIn3( _  
    ByVal Doc As ModelDoc2, _  
    ByVal NewValue As System.Double, _  
    ByVal WhichConfigurations As System.Integer _  
) As System.Integer
```

Visual Basic (Usage)

```
Dim instance As IDimension  
Dim Doc As ModelDoc2  
Dim NewValue As System.Double  
Dim WhichConfigurations As System.Integer  
Dim value As System.Integer
```

```
value = instance.ISetUserValueIn3(Doc, NewValue, WhichConfigurations)
```

C#

```
System.int ISetUserValueIn3(  
    ModelDoc2 Doc,  
    System.double NewValue,  
    System.int WhichConfigurations  
)
```

C++/CLI

```
System.int ISetUserValueIn3(  
& ModelDoc2^ Doc,  
& System.double NewValue,  
& System.int WhichConfigurations  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Doc

[Document](#) whose units you want to use

NewValue

Value for this dimension in the units of the document specified by Doc

WhichConfigurations

Configurations to set the value as defined in [swSetValueInConfiguration_e](#)

Return Value

Success indicator as defined in [swSetValueReturnStatus_e](#)

Visual Basic for Applications (VBA) Syntax

See [Dimension::ISetUserValueIn3](#).

Remarks

The WhichConfigs argument is equivalent to the **Change Parameter** dialog in the SOLIDWORKS Design user interface, which gives the user the option of having the value set in all configurations or the current configuration. If there is one configuration in the part, SOLIDWORKS ignores the argument.

This method allows you to change the value of a read-only dimension. You can use [IDimension::ReadOnly](#) to determine if a dimension is read-only.

See Also

[IDimension Interface](#)
[IDimension Members](#)
[IDimension::GetSystemValue3 Method](#)
[IDimension::GetValue3 Method](#)
[IDimension::IGetSystemValue3 Method](#)
[IDimension::IGetUserValueIn2 Method](#)
[IDimension::IGetSystemValue3 Method](#)
[IDimension::ISetValue3 Method](#)
[IDimension::SetSystemValue2 Method](#)
[IDimension::SetUserValueIn2 Method](#)
[IDimension::SetValue3 Method](#)
[IDimension::GetUserValueIn Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ISetValue3 Method (IDimension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDimension Interface](#) : ISetValue3 Method (IDimension)

Sets the values of the dimensions in the specified configurations.

.NET Syntax

```
Function ISetValue3( _
    ByVal NewValue As System.Double, _
    ByVal WhichConfigurations As System.Integer, _
    ByVal Config\_count As System.Integer, _
    ByRef Config\_names As System.String _
) As System.Integer
```



```
Dim instance As IDimension
Dim NewValue As System.Double
Dim WhichConfigurations As System.Integer
Dim Config_count As System.Integer
Dim Config_names As System.String
Dim value As System.Integer

value = instance.ISetValue3(NewValue, WhichConfigurations, Config_count, Config_names)
```

```
System.int ISetValue3(
    System.double NewValue,
    System.int WhichConfigurations,
    System.int Config\_count,
    ref System.string Config\_names
)
```

```
System.int ISetValue3(
    & System.double NewValue,
    & System.int WhichConfigurations,
    & System.int Config\_count,
    & System.String^% Config\_names
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

NewValue

Value for this dimension in the units of the owning document

WhichConfigurations

- in-process, unmanaged C++: Pointer to an array of configurations in which to set this value as defined in [swSetValueInConfiguration e](#) (see **Remarks**)

- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

Config_count

Number of configurations for which to set dimension values (see **Remarks**)

Config_names

- in-process, unmanaged C++: Pointer to an array of strings of the names of the configurations for which to set dimension values (see **Remarks**)

- VBA, VB.NET, C#, and C++/CLI: Not supported

See [In-process Methods](#) for details about this type of method.

Return Value

Error code as defined in [swSetValueReturnStatus e](#)

Remarks

The Config_count and Config_names arguments are only used if whichConfigurations is set to swSetValue_InSpecificConfigurations. Config_names can contain either a BSTR array or a single BSTR.

See Also

[IDimension Interface](#)
[IDimension Members](#)
[IDimension::GetSystemValue3 Method](#)
[IDimension::GetUserValueIn Method](#)
[IDimension::GetValue3 Method](#)
[IDimension::!GetSystemValue3 Method](#)
[IDimension::!GetValue3 Method](#)
[IDimension::!SetSystemValue3 Method](#)
[IDimension::!SetUserValueIn3 Method](#)
[IDimension::SetSystemValue3 Method](#)
[IDimension::SetUserValueIn2 Method](#)
[IDimension::SetValue3 Method](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IsReference Method (IDimension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDimension Interface](#) : IsReference Method (IDimension)

Gets whether the dimension is a reference dimension.

.NET Syntax

Visual Basic (Declaration)

```
Function IsReference() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IDimension
Dim value As System.Boolean

value = instance.IsReference()
```

C#

```
System.bool IsReference()
```

C++/CLI

```
System.bool IsReference();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if the dimension is a reference dimension, false if not

Visual Basic for Applications (VBA) Syntax

See [Dimension::IsReference](#).

Example

[Get Dimension Values in All Configurations \(VBA\)](#)

Remarks

This method returns false for dimensions in 3D sketches.

See Also

[IDimension Interface](#)
[IDimension Members](#)
[IDrawingDoc::InsertRefDim Method \(\)](#)
[IDisplayDimension::IsReferenceDim Method \(\)](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

Name Property (IDimension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDimension Interface](#) : Name Property (IDimension)

Gets or sets the name of a dimension.

.NET Syntax

Visual Basic (Declaration)

Property Name As System.String

Visual Basic (Usage)



```
Dim instance As IDimension  
Dim value As System.String
```

```
instance.Name = value
```

```
value = instance.Name
```

C#

```
System.string Name {get; set;}
```

C++/CLI

```
property System.String^ Name {  
    System.String^ get();  
    void set ( & System.String^ value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Name of the dimension

Visual Basic for Applications (VBA) Syntax

See [Dimension::Name](#).

Example

[Get Block Information \(VBA\)](#)
[Get Component Via Display Dimension \(VBA\)](#)
[Get Dimension Values in All Configurations \(VBA\)](#)
[Get Dimension Values in Drawing \(VBA\)](#)

Remarks

If setting the name of a dimension, do not include any of the special characters reserved by SOLIDWORKS.

See Also

[IDimension Interface](#)
[IDimension Members](#)
[IDimension::FullName Property](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

ReadOnly Property (IDimension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDimension Interface](#) : ReadOnly Property (IDimension)

Gets or sets the read-only state of a dimension.

.NET Syntax

Visual Basic (Declaration)

```
Property ReadOnly As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IDimension  
Dim value As System.Boolean
```

```
instance.ReadOnly = value
```

```
value = instance.ReadOnly
```

C#

```
System.bool ReadOnly {get; set;}
```

C++/CLI

```
property System.bool ReadOnly {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True if the dimension is read-only, false if it is writable

Visual Basic for Applications (VBA) Syntax

See [Dimension::ReadOnly](#).

Example

[Get Block Information \(VBA\)](#)
[Get Dimension Values in All Configurations \(VBA\)](#)
[Get Dimension Values in Drawing \(VBA\)](#)
[Set Dimensions to Mid-tolerance \(VBA\)](#)

See Also

[IDimension Interface](#)
[IDimension Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

ReferencePoints Property (IDimension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDimension Interface](#) : ReferencePoints Property (IDimension)

Gets or sets the reference points for this dimension.

.NET Syntax

Visual Basic (Declaration)

```
Property ReferencePoints As System.Object
```

Visual Basic (Usage)



```
Dim instance As IDimension  
Dim value As System.Object
```

```
instance.ReferencePoints = value
```

```
value = instance.ReferencePoints
```

C#

```
System.object ReferencePoints {get; set;}
```

C++/CLI

```
property System.Object^ ReferencePoints {  
    System.Object^ get();  
    void set ( & System.Object^ value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Reference points (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [Dimension::ReferencePoints](#).

Remarks

The returned objects, the reference points, are actually [IMathPoint](#) objects.

Reference point 1 and reference point 2 are valid for linear dimensions. Reference point 3 is the vertex for angular dimensions. This property does not distinguish between linear or angular dimensions or other types of dimensions. Three reference points are always returned.

NOTE: When regenerating a [macro feature](#), this property gets and sets the reference points of a display dimension ([IDisplayDimension](#) object). In all other cases, this property gets and sets the reference points of a dimension ([IDimension](#) object).

See Also

[IDimension Interface](#)
[IDimension Members](#)
[IDimension::IGetReferencePoints Method](#)
[IDimension::ISetReferencePoints Method](#)
[IDimension::GetReferencePointsCount Method](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

SetArcEndCondition Method (IDimension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDimension Interface](#) : SetArcEndCondition Method (IDimension)

Sets the end conditions for linear dimensions that end on an arc.

.NET Syntax

Visual Basic (Declaration)

```
Function SetArcEndCondition( _  
    ByVal Index As System.Integer, _  
    ByVal Condition As System.Integer _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IDimension  
Dim Index As System.Integer  
Dim Condition As System.Integer  
Dim value As System.Integer  
  
value = instance.SetArcEndCondition(Index, Condition)
```

C#

```
System.int SetArcEndCondition(  
    System.int Index,  
    System.int Condition  
)
```

C++/CLI

```
System.int SetArcEndCondition(  
& System.int Index,  
& System.int Condition  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Index
Index of the endpoint on which to set the end condition; 1 is the first endpoint, 2 is the second endpoint

Condition
End condition as defined in [swArcEndCondition_e](#)

Return Value

Indicates the success or failure of this method:

0	Command was successful; the arc end condition was set
-1	Command failed for an unknown reason; the arc end condition was not set
-2	Index parameter is invalid
-3	Condition parameter is invalid
-4	Endpoint 1 is not related to an arc
-5	Endpoint 2 is not related to an arc

 Visual Basic for Applications (VBA) Syntax

See [Dimension::SetArcEndCondition](#).

 Remarks

Linear dimensions measure the distance from one point to another. If one or both of those points is on an arc, the point can be changed to the center point of the arc, the nearest point on the arc, or the furthest point on the arc. The arc end condition describes which point to use.

Use [IDimension::GetArcEndCondition](#) to get the arc end conditions.

To see the effects of changing the arc endpoint conditions, use [IModelDoc2::GraphicsRedraw2](#) to redraw the graphics window.

 See Also

[IDimension Interface](#)
[IDimension Members](#)
[IDimension::GetArcEndCondition Method](#)

 Availability

SOLIDWORKS 99, datecode 1999207

SOLIDWORKS API Help [Send comments on this topic.](#)

SetSystemValue2 Method (IDimension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDimension Interface](#) : SetSystemValue2 Method (IDimension)

Obsolete. Superseded by [IDimension::SetSystemValue3](#).

 .NET Syntax

Visual Basic (Declaration)

```
Function SetSystemValue2( _  
    ByVal NewValue As System.Double, _  
    ByVal WhichConfigurations As System.Integer _  
) As System.Integer
```

Visual Basic (Usage)

```
Dim instance As IDimension  
Dim NewValue As System.Double  
Dim WhichConfigurations As System.Integer  
Dim value As System.Integer
```

```
value = instance.SetSystemValue2(NewValue, WhichConfigurations)
```

C#

```
System.int SetSystemValue2(
    System.double NewValue,
    System.int WhichConfigurations
)
```

C++/CLI

```
System.int SetSystemValue2(
    & System.double NewValue,
    & System.int WhichConfigurations
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

NewValue

WhichConfigurations

Visual Basic for Applications (VBA) Syntax

See [Dimension::SetSystemValue2](#).

See Also

[IDimension Interface](#)
[IDimension Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

SetSystemValue3 Method (IDimension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDimension Interface](#) : SetSystemValue3 Method (IDimension)

Sets the value of this dimension in system units (meters) in the specified configuration.

.NET Syntax

Visual Basic (Declaration)

```
Function SetSystemValue3( _
    ByVal NewValue As System.Double, _
    ByVal WhichConfigurations As System.Integer, _
    ByVal Config\_names As System.Object _
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IDimension
Dim NewValue As System.Double
Dim WhichConfigurations As System.Integer
Dim Config_names As System.Object
Dim value As System.Integer

value = instance.SetSystemValue3(NewValue, WhichConfigurations, Config_names)
```

C#

```
System.int SetSystemValue3(
    System.double NewValue,
    System.int WhichConfigurations,
    System.object Config\_names
)
```

C++/CLI

```
System.int SetSystemValue3(
    & System.double NewValue,
```

```
& System.int WhichConfigurations,  
& System.Object^ Config_names  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

NewValue

Dimension value in meters

WhichConfigurations

Configuration in which to set this value as defined in [swSetValueInConfiguration_e](#) (see **Remarks**)

Config_names

Names of the configurations (see **Remarks**)

Return Value

Success indicator value as defined in [swSetValueReturnStatus_e](#)

Visual Basic for Applications (VBA) Syntax

See [Dimension::SetSystemValue3](#).

Example

[Change Dimensions of Gear Mate \(VBA\)](#)
[Modify Plane By Changing System Value \(VBA\)](#)
[Set Dimensions to Mid-tolerance \(VBA\)](#)
[Recalculate Bounding Box \(C#\)](#)
[Recalculate Bounding Box \(VB.NET\)](#)
[Recalculate Bounding Box \(VBA\)](#)

Remarks

The WhichConfigurations argument is equivalent to the **Change Parameter** dialog in the SOLIDWORKS Design user interface, which gives the user the option of having the value set in all configurations or the current configuration. If there is one configuration in the part, SOLIDWORKS ignores this argument.

Config_names argument is only used if WhichConfigurations is set to swSetValue_InSpecificConfigurations and can contain either a BSTR array or a single BSTR.

This method allows you to change the value of a read-only dimension. You can use [IDimension::ReadOnly](#) to determine if a dimension is read-only.

See Also

[IDimension Interface](#)
[IDimension Members](#)
[IDimension::GetSystemValue3 Method](#)
[IDimension::GetUserValueIn Method](#)
[IDimension::GetValue3 Method](#)
[IDimension::!GetSystemValue3 Method](#)
[IDimension::!GetUserValueIn2 Method](#)
[IDimension::!GetValue3 Method](#)
[IDimension::!SetSystemValue3 Method](#)
[IDimension::!SetUserValueIn3 Method](#)
[IDimension::!SetValue3 Method](#)
[IDimension::SetUserValueIn2 Method](#)
[IDimension::SetValue3 Method](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

SetToleranceFitValues Method (IDimension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDimension Interface](#) : SetToleranceFitValues Method (IDimension)

Obsolete. Superseded by [IDimensionTolerance::SetFitValues](#).

.NET Syntax

Visual Basic (Declaration)

```
Function SetToleranceFitValues( _  
    ByVal NewLValue As System.String, _  
    ByVal NewUValue As System.String _  
    ) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IDimension  
Dim NewLValue As System.String  
Dim NewUValue As System.String  
Dim value As System.Boolean  
  
value = instance.SetToleranceFitValues(NewLValue, NewUValue)
```

C#

```
System.bool SetToleranceFitValues(  
    System.string NewLValue,  
    System.string NewUValue  
)
```

C++/CLI

```
System.bool SetToleranceFitValues(  
& System.String^ NewLValue,  
& System.String^ NewUValue  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

NewLValue

NewUValue

Visual Basic for Applications (VBA) Syntax

See [Dimension::SetToleranceFitValues](#).

See Also

[IDimension Interface](#)
[IDimension Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

SetToleranceFontInfo Method (IDimension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDimension Interface](#) : SetToleranceFontInfo Method (IDimension)

Obsolete. Superseded by [IDimensionTolerance::SetFont](#).

.NET Syntax

Visual Basic (Declaration)

```
Function SetToleranceFontInfo( _  
    ByVal UseFontScale As System.Integer, _  
    ByVal TolScale As System.Double, _  
    ByVal TolHeight As System.Double _  
    ) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IDimension  
Dim UseFontScale As System.Integer  
Dim TolScale As System.Double
```

```
Dim TolHeight As System.Double
Dim value As System.Boolean

value = instance.SetToleranceFontInfo(UseFontScale, TolScale, TolHeight)
```

C#

```
System.bool SetToleranceFontInfo(
    System.int UseFontScale,
    System.double TolScale,
    System.double TolHeight
)
```

C++/CLI

```
System.bool SetToleranceFontInfo(
    & System.int UseFontScale,
    & System.double TolScale,
    & System.double TolHeight
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

UseFontScale

TolScale

TolHeight

Visual Basic for Applications (VBA) Syntax

See [Dimension::SetToleranceFontInfo](#).

See Also

[IDimension Interface](#)
[IDimension Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

SetToleranceType Method (IDimension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDimension Interface](#) : SetToleranceType Method (IDimension)

Obsolete. Superseded by [IDimensionTolerance::Type](#).

.NET Syntax

Visual Basic (Declaration)

```
Function SetToleranceType( _
    ByVal NewType As System.Integer _
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IDimension
Dim NewType As System.Integer
Dim value As System.Boolean

value = instance.SetToleranceType(NewType)
```

C#

```
System.bool SetToleranceType(
    System.int NewType
)
```

C++/CLI

```
System.bool SetToleranceType(  
& System.int NewType  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

NewType

Visual Basic for Applications (VBA) Syntax

See [Dimension::SetToleranceType](#).

See Also

[IDimension Interface](#)
[IDimension Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

SetToleranceValues Method (IDimension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDimension Interface](#) : SetToleranceValues Method (IDimension)

Obsolete. Superseded by [IDimensionTolerance::SetValues](#).

.NET Syntax

Visual Basic (Declaration)

```
Function SetToleranceValues( _  
    ByVal TolMin As System.Double, _  
    ByVal TolMax As System.Double _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IDimension  
Dim TolMin As System.Double  
Dim TolMax As System.Double  
Dim value As System.Boolean
```

```
value = instance.SetToleranceValues(TolMin, TolMax)
```

C#

```
System.bool SetToleranceValues(  
    System.double TolMin,  
    System.double TolMax  
)
```

C++/CLI

```
System.bool SetToleranceValues(  
& System.double TolMin,  
& System.double TolMax  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

TolMin

TolMax

Visual Basic for Applications (VBA) Syntax

See [Dimension::SetToleranceValues](#).

See Also

[IDimension Interface](#)
[IDimension Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

SetUserValueIn Method (IDimension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDimension Interface](#) : SetUserValueIn Method (IDimension)

Obsolete. Superseded by [IDimension::SetUserValueIn2](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub SetUserValueIn( _  
    ByVal Doc As System.Object, _  
    ByVal NewValue As System.Double _  
)
```

Visual Basic (Usage)



```
Dim instance As IDimension  
Dim Doc As System.Object  
Dim NewValue As System.Double  
  
instance.SetUserValueIn(Doc, NewValue)
```

C#

```
void SetUserValueIn(  
    System.object Doc,  
    System.double NewValue  
)
```

C++/CLI

```
void SetUserValueIn(  
& System.Object^ Doc,  
& System.double NewValue  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Doc

NewValue

Visual Basic for Applications (VBA) Syntax

See [Dimension::SetUserValueIn](#).

See Also

[IDimension Interface](#)
[IDimension Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

SetUserValueIn2 Method (IDimension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDimension Interface](#) : SetUserValueIn2 Method (IDimension)

Sets the value of this dimension in the units of the specified document.

Visual Basic Syntax

Visual Basic (Declaration)

```
Function SetUserValueIn2( _  
    ByVal Doc As System.Object, _  
    ByVal NewValue As System.Double, _  
    ByVal WhichConfigurations As System.Integer _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IDimension  
Dim Doc As System.Object  
Dim NewValue As System.Double  
Dim WhichConfigurations As System.Integer  
Dim value As System.Integer  
  
value = instance.SetUserValueIn2(Doc, NewValue, WhichConfigurations)
```

C#

```
System.int SetUserValueIn2(  
    System.object Doc,  
    System.double NewValue,  
    System.int WhichConfigurations  
)
```

C++/CLI

```
System.int SetUserValueIn2(  
& System.Object^ Doc,  
& System.double NewValue,  
& System.int WhichConfigurations  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Doc

Document whose units you want to use

NewValue

Value for this dimension in the units of the Doc argument

WhichConfigurations

Configurations to set this value in as defined in [swSetValueInConfiguration_e](#)

Return Value

Success indicator as defined in [swSetValueReturnStatus_e](#)

Visual Basic for Applications (VBA) Syntax

See [Dimension::SetUserValueIn2](#).

Remarks

The WhichConfigs argument is equivalent to the **Change Parameter** dialog in the SOLIDWORKS Design user interface, which gives the user the option of having the value set in all configurations or the current configuration. If there is one configuration in the part, SOLIDWORKS ignores this argument.

This method allows you to change the value of a read-only dimension. You can use [IDimension::ReadOnly](#) to determine if a dimension is read-only.

See Also

[IDimension Interface](#)
[IDimension Members](#)
[IDimension::GetSystemValue3 Method](#)
[IDimension::GetUserValueIn Method](#)
[IDimension::GetValue3 Method](#)

[IDimension::IGetSystemValue3 Method](#)
[IDimension::IGetUserValueIn2 Method](#)
[IDimension::IGetValue3 Method](#)
[IDimension::ISetSystemValue3 Method](#)
[IDimension::ISetUserValueIn3 Method](#)
[IDimension::ISetValue3 Method](#)
[IDimension::SetSystemValue3 Method](#)
[IDimension::SetValue3 Method](#)

Availability

SOLIDWORKS 98Plus, datecode 1999005

SOLIDWORKS API Help

[Send comments on this topic.](#)

SetValue2 Method (IDimension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDimension Interface](#) : SetValue2 Method (IDimension)

Obsolete. Superseded by [IDimension::SetValue3](#).

.NET Syntax

Visual Basic (Declaration)

```
Function SetValue2( _  
    ByVal NewValue As System.Double, _  
    ByVal WhichConfigurations As System.Integer _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IDimension  
Dim NewValue As System.Double  
Dim WhichConfigurations As System.Integer  
Dim value As System.Integer  
  
value = instance.SetValue2(NewValue, WhichConfigurations)
```

C#

```
System.int SetValue2(  
    System.double NewValue,  
    System.int WhichConfigurations  
)
```

C++/CLI

```
System.int SetValue2(  
& System.double NewValue,  
& System.int WhichConfigurations  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

NewValue

WhichConfigurations

Visual Basic for Applications (VBA) Syntax

See [Dimension::SetValue2](#).

See Also

[IDimension Interface](#)
[IDimension Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

SetValue3 Method (IDimension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDimension Interface](#) : SetValue3 Method (IDimension)

Sets the values of the dimensions in the specified configurations.

.NET Syntax

Visual Basic (Declaration)

```
Function SetValue3( _  
    ByVal NewValue As System.Double, _  
    ByVal WhichConfigurations As System.Integer, _  
    ByVal Config\_names As System.Object _  
    ) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IDimension  
Dim NewValue As System.Double  
Dim WhichConfigurations As System.Integer  
Dim Config_names As System.Object  
Dim value As System.Integer  
  
value = instance.SetValue3(NewValue, WhichConfigurations, Config_names)
```

C#

```
System.int SetValue3(  
    System.double NewValue,  
    System.int WhichConfigurations,  
    System.object Config\_names  
)
```

C++/CLI

```
System.int SetValue3(  
& System.double NewValue,  
& System.int WhichConfigurations,  
& System.Object^ Config\_names  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

NewValue

Value for this dimension in the units of owning document

WhichConfigurations

Configurations in which to set this value as defined in [swSetValueInConfiguration e](#) (see **Remarks**)

Config_names

Names of the configurations for which to set dimension values (see **Remarks**)

Return Value

Error code as defined in [swSetValueReturnStatus e](#)

Visual Basic for Applications (VBA) Syntax

See [Dimension::SetValue3](#).

Remarks

The Config_names argument is only used if WhichConfigurations is set to swSetValue_InSpecificConfigurations and can contain either a BSTR array or a single BSTR.

See Also

[IDimension Interface](#)
[IDimension Members](#)
[IDimension::GetSystemValue3 Method](#)
[IDimension::GetUserValueIn Method](#)
[IDimension::GetValue3 Method](#)

[IDimension::IGetSystemValue3 Method](#)
[IDimension::IGetUserValueIn2 Method](#)
[IDimension::IGetValue3 Method](#)
[IDimension::ISetSystemValue3 Method](#)
[IDimension::ISetUserValueIn3 Method](#)
[IDimension::ISetValue3 Method](#)
[IDimension::SetSystemValue3 Method](#)
[IDimension::SetUserValueIn2 Method](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

SystemValue Property (IDimension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDimension Interface](#) : SystemValue Property (IDimension)

Obsolete. Superseded by [IDimension::GetSystemValue3](#), [IDimension::IGetSystemValue3](#), [IDimension::SetSystemValue3](#), and [IDimension::ISetSystemValue3](#).

.NET Syntax

Visual Basic (Declaration)

```
Property SystemValue As System.Double
```

Visual Basic (Usage)



```
Dim instance As IDimension  
Dim value As System.Double  
  
instance.SystemValue = value  
value = instance.SystemValue
```

C#

```
System.double SystemValue {get; set;}
```

C++/CLI

```
property System.double SystemValue {  
    System.double get();  
    void set ( & System.double value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [Dimension::SystemValue](#).

See Also

[IDimension Interface](#)
[IDimension Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

Tolerance Property (IDimension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDimension Interface](#) : Tolerance Property (IDimension)

Gets the [IDimensionTolerance](#) object.

.NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property Tolerance As DimensionTolerance
```

```
Dim instance As IDimension
Dim value As DimensionTolerance

value = instance.Tolerance
```

C#

```
DimensionTolerance Tolerance {get;}
```

C++/CLI

```
property DimensionTolerance^ Tolerance {
    DimensionTolerance^ get();
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Pointer to the [IDimensionTolerance](#) object

Visual Basic for Applications (VBA) Syntax

See [Dimension::Tolerance](#).

Example

[Get Dimension Tolerance \(VBA\)](#)
[Get Dimension Tolerance \(VB.NET\)](#)
[Get Dimension Tolerance \(C#\)](#)

See Also

[IDimension Interface](#)
[IDimension Members](#)

Availability

SOLIDWORKS 2006 FCS, Revision Number 14.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

Value Property (IDimension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IDimension Interface](#) : Value Property (IDimension)

Obsolete. Superseded by [IDimension::GetValue3](#), [IDimension::!GetValue3](#), [IDimension::!SetValue3](#), and [IDimension::SetValue3](#).

.NET Syntax

Visual Basic (Declaration)

```
Property Value As System.Double
```

Visual Basic (Usage)

```
Dim instance As IDimension
Dim value As System.Double

instance.Value = value

value = instance.Value
```

C#

```
System.double Value {get; set;}
```

C++/CLI

```
property System.double Value {  
    System.double get();  
    void set ( &    System.double value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [Dimension::Value](#).

See Also

[IDimension Interface](#)
[IDimension Members](#)